

Project Title: Image Classification using Binarized Neural Networks (BNN) on FPGA with PYNQ Board

Environment: Jupyter Notebook, Python 3.6, PYNQ Overlay, CFFI Library

1. Objective

The goal of this lab is to load a custom FPGA bitstream, interact with the BNN hardware accelerator, and perform inference for image classification tasks using a PYNQ board.

2. Initial Setup

We began by importing required Python libraries and setting up the FPGA overlay:

```
from pynq import Overlay
import cffi
import bnn
```

```
ROOT = "/opt/python3.6/lib/python3.6/site-packages/bnn/"
```

```
overlay_path = "/home/xilinx/jupyter_notebooks/FPGA-Project-BNN
1/FPGA-Project-BNN/procsys_wrapper/procsys_wrapper.bit"
ol = Overlay(overlay_path)
ol.download()
```

```
ffi = cffi.FFI()
ffi.cdef("""
    void load_parameters(const char* path);
    unsigned int inference(const char* path, unsigned int results[64],
int number_class, float *usecPerImage);
    unsigned int* inference_multiple(const char* path, int
number_class, int image_number, float *usecPerImage, unsigned int
enable_detail);
    void free_results(unsigned int * result);
```

```
    void deinit();  
    """)
```

Result: ♦ Successful overlay loading and CFFI setup.

3. First Major Error: Syntax Error When Loading Parameters

Initial attempt to define a load_params function:

```
def load_params(/home/xilinx/jupyter_notebooks/BNN-  
PYNQ/bnn/params/mnist/lfcW1A2/):  
    params_path_c = ffi.new("char[]", params_path.encode('utf-8'))  
    ol.load_parameters(params_path_c)
```

Error Encountered:

SyntaxError: invalid syntax

Analysis:

- Path was incorrectly placed directly in the function definition.

Action Taken:

- Corrected function definition:

```
def load_params(params_path):  
    params_path_c = ffi.new("char[]", params_path.encode('utf-8'))  
    ol.load_parameters(params_path_c)
```

```
load_params("/home/xilinx/jupyter_notebooks/BNN-  
PYNQ/bnn/params/mnist/lfcW1A2/")
```

4. Second Major Error: Missing Function in Overlay

Upon running the corrected function:

Error Encountered:

AttributeError: Could not find IP or hierarchy load_parameters in overlay

Analysis:

- load_parameters not exposed by the FPGA bitstream.

Action Taken:

- Decided to manually load the shared object library libbnn.so instead.

5. Third Major Error: Shared Object Library Missing

Attempted loading of libbnn.so:

```
lib = ffi.dlopen("/opt/python3.6/lib/python3.6/site-packages/bnn/libbnn.so")
```

Error Encountered:

OSError: cannot load library '/opt/python3.6/lib/python3.6/site-packages/bnn/libbnn.so': No such file or directory

Analysis:

- The shared object libbnn.so was missing.

Action Taken:

- Planned to search for libbnn.so using terminal commands:

```
find /opt -name "libbnn.so"
```

or globally:

```
find / -name "libbnn.so"
```

If missing, a reinstall of the bnn package or rebuilding the shared object was considered necessary.

6. Summary of Actions and Errors Chronologically

Step	Action	Outcome
1	Imported libraries and loaded overlay	✓ Successful
2	Defined load_params incorrectly	✗ Syntax Error
3	Corrected load_params function	✓ Syntax Fixed
4	Called load_parameters from overlay	✗ AttributeError (not found in overlay)
5	Attempted to load libbnn.so	✗ OSError (file missing)
6	Planned next step: search or reinstall bnn	🕒 In Progress

7. Conclusions and Recommendations

- **Overlay Issue:** The bitstream used does not expose the expected IP block.
- **Software Issue:** The required shared object libbnn.so is missing.

Next Actions:

1. Search if libbnn.so exists anywhere on the system.
2. If not found, reinstall the bnn package or rebuild it.
3. Verify that the FPGA bitstream includes the necessary IPs.
4. Optionally review the PYNQ BNN repository setup instructions.

Final Note:

The encountered problems appear to be both hardware (overlay bitstream mismatch) and software (missing libraries). Resolving both issues is essential to achieve successful BNN inference.

